Evaluating AI Agent Teams with Decentralized Information

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Motivation and Goal

AI agent teams must often combine information held by different members. In a financial review, one agent may receive a cash balance, another a restriction on using that cash, and a third a repayment deadline. They share the objective of deciding whether a payment is feasible. Each agent must find the relevant facts in its own records and communicate what the others need. Strong performance by a single agent given every fact does not establish whether it can contribute effectively in this setting.

Team decision theory and decentralized control provide a principled starting point [1,2]. Decentralized information refers to agents making decisions from information that differs across members. The information structure specifies which observations, past actions, and received messages each agent can use at each decision step. Communication is itself a decision that changes what other members can observe. We will use this perspective to evaluate AI agent teams, holding the decision problem fixed while changing information access and communication. Arena already measures how model choices affect real work and its cost [3,4]. Our contribution will connect these comparisons to the evidence agents retrieve, exchange, and use.

How can we evaluate and improve AI agent teams when the information needed for a decision is decentralized?

Thrust 1 Evaluating Decisions under Decentralized Information

Construct decisions with checked evidence. Building on our ongoing prototype, we will construct tasks from financial documents and software repositories. For each task, we will identify pivotal facts that together support a reference decision. Removing any selected fact must leave that decision unsupported under our checks. We will examine alternative evidence and independently review whether an agent's local records, together with shared information, already support the full answer. These checks make changes in information access meaningful rather than merely changing document length.

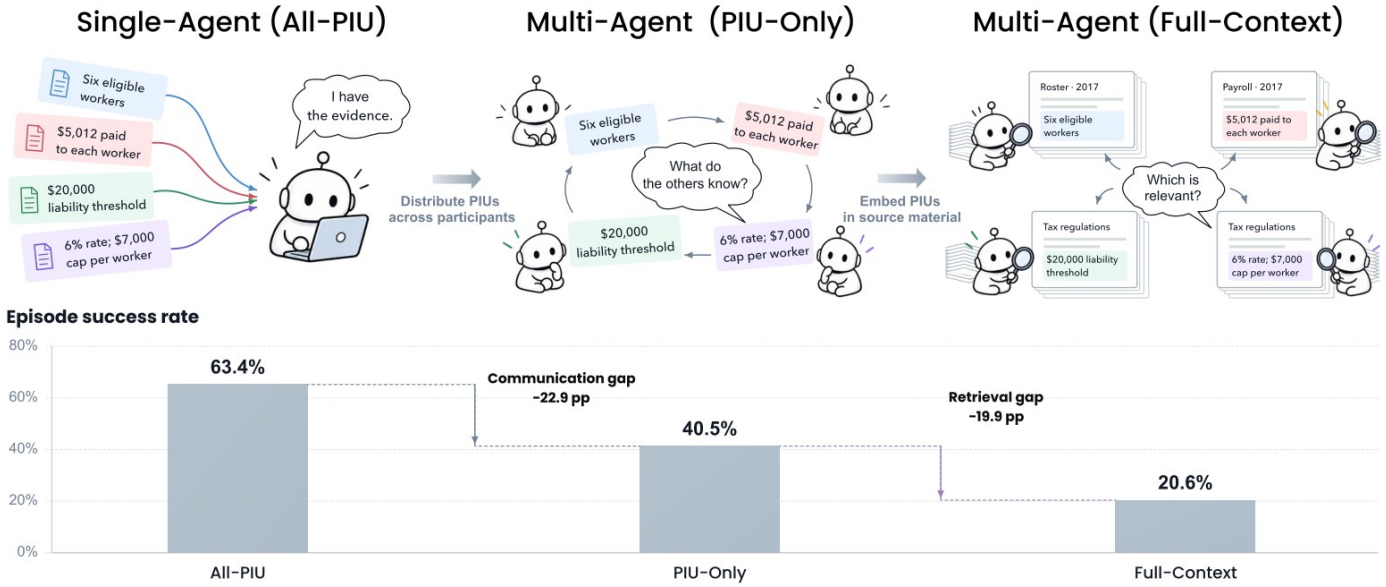


Figure 1. Preliminary success rates across eight models and twelve source corpora. A PIU is a pivotal information unit, or checked piece of evidence. Success is the fraction of task runs meeting the scoring rule. Rates average models within each corpus, then weight corpora equally. The two AI agent team conditions share communication rules and resource limits.

Separate finding information from communicating it. We will compare one agent receiving all pivotal facts, an AI agent team receiving those facts separately, and the same team receiving source documents containing its assigned facts. The communication gap is the drop in success when the facts are split across agents. The retrieval gap is the further drop when they must extract those facts from documents (Figure 1). On selected tasks, we will vary whether a fact goes to one agent or all agents, and whether most facts go to one member.

Evaluate communication rules and model choices. Members will speak in a fixed order, report without seeing one another's messages, or respond to a coordinator that selects whom to query and makes the final decision. This will test whether a communication method's advantage changes with the information structure. We will compare six models on the same tasks, including replacing a participant or coordinator while keeping its partners fixed. Comparisons will match total limits on tokens read and generated, and report actual team cost, completion, and accuracy. Completion means returning a valid decision within the resource limit. Repeated runs and uncertainty intervals will assess model rankings under each tested information condition.

Thrust 2 Diagnosing Failures and Validating Improvements

Diagnose where relevant information is lost. We will check which pivotal facts appear in each agent's extracted evidence and messages, and whether the final decision correctly accounts for them. To test specific failures, we will hold a conversation fixed and regenerate only its final decision. We will add a correct inference supported by facts already in the conversation, without supplying the answer, or remove a later statement that relies on an earlier error while retaining the original error. Using the same final-decision model will test what these edits change for those conversations. Separately, we will broaden narrow coordinator questions while keeping participants, speaking order, and response limits fixed, and measure whether more relevant evidence is reported.

Use training to validate the evaluation. Training lets us test whether a higher final score reflects better collaboration. Building on initial trials, we will use reinforcement learning with a reward for the final decision. With a coordinator, we will compare training participants, the coordinator, and both together. Without one, we will train a shared participant model to choose messages, the next speaker, and when to decide. We will test whether gains in the final score coincide with better information coverage and use, without separately rewarding a prescribed communication pattern.

Test whether improvements carry over. We will conduct three independent training runs and evaluate on new source documents, team sizes, and allocations of facts. Preliminary experiments show that better participation need not improve evidence transmission on a new corpus, so both will be measured. The same model will generate final decisions from conversations produced before and after training, including valid messages from runs that ended without a decision. This will test what information the conversations support alongside end-to-end performance. We will also vary message limits to determine whether gains persist under constrained communication.

Evaluation Plan and Outputs

During months 1–3, we will independently audit approximately 200 finance and software tasks and establish their evidence and scoring rules. Months 4–6 will compare the six models under controlled information access and communication rules. Months 7–9 will test communication changes and the focused training study. Months 10–12 will evaluate transfer and release the results. Protocol and budget variations will use selected task subsets. Training will use one pretrained base model, with separate copies for participant and coordinator roles. Related tasks from the same document or repository will remain together when separating training and evaluation data. Final decisions will be checked against reference answers, and code changes will use repository tests. Human review will check a sample of automated scores. We will release permitted tasks, evaluation code, and annotated conversations, with accuracy, uncertainty, and total team cost. Collaboration with Arena researchers will help align the tasks and reporting with actual use. Private Arena data is optional.

We request $50,000 over 12 months: $30,000 for partial graduate research support and benefits, $12,000 for model runs and computing, $5,000 for independent checks, and $3,000 for releasing code and sharing results. Final allocations will follow institutional gift policy.

Selected References

[1] Ho and Chu. Team Decision Theory and Information Structures in Optimal Control Problems—Part I. IEEE TAC, 17(1):15–22, 1972.

[2] Nayyar, Mahajan, and Teneketzis. Decentralized Stochastic Control with Partial History Sharing: A Common Information Approach. IEEE TAC, 58(7):1644–1658, 2013.

[3] Arena. Agent Arena: Causal Evaluation of Agents in the Real World, 2026. arena.ai/blog/agent-arena-methodology.

[4] Arena. Agent Leaderboard Improvements: Categories & Task Cost, 2026. arena.ai/blog/agent-categories-and-cost.